MIL-R-21017A(SHIPS)
23 July 1965
SUPERSEDING
MIL-R-21017(SHIPS)
12 August 1957
(See 6.3)

MILITARY SPECIFICATION

RADIACMETER, ALARM, PORTABLE

1. SCOPE

1.1 This specification covers a portable, dose indicating alarm radiacmeter. This equipment shall detect and indicate the integrated quantity of X or gamma radiation or both to which it has been exposed, and shall actuate an aural alarm when a predetermined dose is reached.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-B-18 - Batteries, Dry.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-R-22732 - Reliability Requirements for Shipboard and Ground Electronic Equipment.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Ave. at 33rd. Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Preproduction sample. - Prior to beginning production a sample shall be tested as specified in 4.3.

3.2 General. - The equipment shall be in accordance with all provisions of MIL-E-16400 except as otherwise specified herein.

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3.3 Reliability. - Reliability requirements shall be as follows:

3.3.1 The contractor shall conduct the reliability assurance program that is applicable to a pre-production and production model.

3.3.2 Level. - The reliability requirement for this equipment shall be 5000 hours, mean-time-between-failures (MTBF).

3.3.3 Reliability evaluation. - Reliability evaluation shall be performed in accordance with the procedures for "Reliability Assurance by Analysis and Prediction", as specified in MIL-R-22732.

3.3.4 Testing. - Quantitative reliability requirements shall be demonstrated as specified in 4.2.2.

3.4 Design requirements. - Design requirements shall be as follows:

3.4.1 Composition. - The instrument shall consist of a detector and associated circuitry, dose measuring circuits, digital dose register, audible sound and batteries. Batteries shall not be supplied with the equipment.

3.4.1.1 Construction (modular). - The requirements for modular construction are not applicable.

3.4.2 Environmental conditions. - The equipment shall be capable of operation under only the following conditions as specified in MIL-E-16400 or as modified herein:

(a) Temperature, class 3.
(b) Humidity.

3.4.3 Electrical design. - The equipment shall be operated by two BA-30 batteries in accordance with MIL-B-18 and capable of 40 hours of continuous normal operation in a field 50 roentgens per hour r/hr. The batteries shall be in a separate watertight compartment and the batteries readily replaceable.

3.4.3.1 External connectors shall be used in accordance with MIL-E-16400.

3.4.3.2 Blown fuse indicators are not required.

3.4.4 Maintenance. - Test points and test features are required. The test points shall be accessible.

3.4.5 Mechanical. - Mechanical requirements shall be as follows:

3.4.5.1 Shock, vibration and inclination requirements are applicable to this equipment.

3.4.5.1.1 Shock requirements shall be for grade B, class 1, lightweight, type A in accordance with MIL-S-901.

3.4.5.2 The degree of enclosure shall be watertight in accordance with MIL-STD-108.

3.4.6 Controls, indicators and panel layout. - Controls, indicators, and panel layout requirements shall be as follows:

(a) Tactile identification of controls is not required.
(b) Locking devices on controls are not required.
(c) Elapsed time meters are not required.

3.4.7 Instruction plates are not required but required technical information shall be in the technical manual.

3.4.7.1 In addition to the requirements of the technical manual specification, the contractor shall determine the parts that can be replaced without effecting the calibration, and list all such parts under the section entitled "Maintenance".

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3.5 Performance and detail requirements. -

3.5.1 Description. - This equipment shall be a recycling ion chamber type integrator capable of detecting and indicating the total quantity of X or gamma radiation or both to which it has been exposed. Accumulated dose shall be visually indicated by a digital counter. An aural alarm shall be included to indicate when a predetermined dose setting on the counter has been reached.

3.5.1.1 Range. - There shall be a single range of dose indicated from 0 to 1000 roentgens, in increments of 100 milliroentgens. Integration shall be accomplished by recycling the detector.

3.5.1.2 Energy dependency. - The energy dependency of the equipment shall be within plus or minus 20 percent for gamma ray energies from 0.08 MEV to 1.3 MEV and for radiation rates up to 1000 r/hr. and down to radiation rates equivalent to 0.1 r/hr, and for a total dose of any multiple of the incremental range. The reference energy shall be Co^{60}.

3.5.1.3 Dose accuracy. - In the calibrating direction the integrated dose shall be within ± 20 percent of true dose using Co^{60} as the reference standard. The three points to be tested are 20 percent, 50 percent, and 80 percent of full scale.

3.5.1.4 Size. - The size of the equipment shall not exceed 68 cubic inches and shall be no greater than 8-1/2 inches in length overall.

3.5.1.5 Weight. - The weight of the equipment shall not exceed 3-1/3 pounds.

3.5.1.6 Temperature. - All temperature requirements shall be exclusive of batteries.

3.5.1.7 High radiation rate independence. - At a dose rate of 10,000 r/hr., the recycling rate shall be not less than 5 per second.

3.5.1.8 Leakage. - The leakage shall not exceed one increment per 12 hours.

3.5.2 Equipment composition. - The equipment shall consist of everything necessary to meet the requirements of this specification and shall include:

3.5.2.1 Detector unit. - The detector shall be an ion chamber or equivalent detector. Detection and integration of an incremental dose by the discharge of the detector shall through associated electronic circuitry, operate the digital counter, thus adding that increment to the indicated dose. The detector shall then be recharged automatically, the process shall be repeated, and each successive increment shall be recorded on the counter.

3.5.2.2 Visual indicator. - The visual indicator shall be a digital counter operated by a relay. Detection and integration of an incremental dose shall be indicated by operation of the counter as specified in 3.5.2.1. The counter shall permit the setting of a predetermined alarm dose at 1 roentgen and 10 roentgens and this alarm shall not sound for other values of dose. Detection and recording of the predetermined alarm dose shall operate an aural alarm. The counter shall continue to operate after the predetermined value has been recorded.

3.5.2.3 Aural alarm. - An aural alarm shall be built into the equipment and shall be as loud as is practical with the size and weight of the equipment.

3.5.2.4 Aural indication. - Each cycling operation shall momentarily actuate the built-in alarm.

3.5.2.5 Remote readout plug. - The equipment shall have an outlet to permit remote indications.

3.5.2.6 Control switch. - An externally operated control switch shall be provided with the following positions:

(a) "OFF" - Equipment inoperative.
(b) "OPERATING" - Normal operation.
(c) "OPERATING-ALARM OFF" - Normal operation with the exception that alarm will not operate.

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3.5.2.7 Illumination switch. - An externally operated switch shall be provided for illumination of the visual indicator.

3.6 Painting. - The type enamel (air-drying speed) shall be optional. The color shall be as specified for portable shipboard equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system. - The contractor shall provide and maintain and effective inspection and quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the cognizant Government inspector for system approval prior to preproduction inspection. Any changes to the approved quality assurance system which might affect the degree of assurance required by this specification or other applicable documents must be submitted to the cognizant inspector and approved in writing prior to use.

4.1.2 Government Verification. - All quality assurance operations performed by the contractor will be subject to Government verification at anytime. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, and (b) Government product inspection to measure quality of product to be offered for acceptance. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.2 General inspection. - The general inspection of the equipment shall consist of the inspection of MIL-E-16400.

4.2.1 Test, test equipment and test procedures. - A list of proposed tests, test equipment and test procedures shall be furnished 60 days prior to setting up test equipment for approval by the bureau or agency concerned unless otherwise specified in the contract or order.

4.2.2 Reliability testing. - Reliability testing shall be performed in accordance with test level 1 of MIL-R-22732, except that the on-off cycle input voltage variation is not required.

4.2.2.1 Accelerated life tests specified in MIL-E-16400 will not be required for preproduction equipment. In lieu of this test, reliability tests as specified herein will be required.

4.2.2.2 Demonstration of the required reliability level of 5000 hours (MTBF) shall be accomplished by demonstrating a MTBF of 300 hours with 90 percent confidence. The test time on each equipment shall be no more than 700 hours. Accept-reject criteria shall be in accordance with MIL-R-22732.

4.2.2.3 Sampling for the reliability tests shall be on the basis of one unit for every 25 units.

4.2.2.4 The performance parameters to be monitored shall be as specified in 3.5.1.3.

4.2.3 Detector assembly. - Each detector assembly shall be temperature cycled and tested for gas leaks before being installed in an equipment. The tests and temperature cycle shall be as follows:

(a) In a circuit with the same characteristics as those of the production radiacmeter, the detector units response to Cs^{137} or Co^{60} shall be determined.
(b) Decrease temperature to -62°C. ± 2°C. and hold for at least 72 hours.
(c) Increase temperature to + 75°C. ± 2°C. and hold for at least 72 hours.
(d) Check the detector assembly response in accordance with 4.2.3(a). The response shall not vary more than 5 percent from the reading taken as specified in 4.2.3(a).

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(e) All detector units shall be stored for a period of 30 days and again checked for response in accordance with 4. 2. 3(a). The response shall not vary more than 5 percent from the reading taken as specified in 4. 2. 3(a).

4. 3 Preproduction inspection. - The preproduction inspection shall consist of the test specified in MIL-E-16400, except that the following tests shall not be performed:

(a) Supply line voltage and frequency.
(b) Water cooling.
(c) Power.
(d) Control and control circuits.
(e) Accelerated life test.

4. 3. 1 In addition to the tests of 4. 3, the following tests shall be performed:

(a) Energy dependency (see 4. 8. 1).
(b) Dose accuracy (see 4. 8. 2).
(c) Leakage (see 4. 8. 4).
(d) High radiation rate independence (see 4. 8. 3).

4. 4 Production inspection shall be in accordance with MIL-E-16400 and the following:

(a) Accuracy abbreviated (see 4. 8. 2. 1).
(b) Leakage (see 4. 8. 4).

4. 5 Production control inspection. - The production control inspection shall consist of the following tests of MIL-E-16400:

(a) Weights and dimensions.
(b) Heat test.
(c) Enclosure.
(d) Reliability.

4. 5. 1 In addition to the tests specified in 4. 5, dose accuracy test (see 4. 8. 2) shall be performed.

4. 5. 2 Sampling plan. - The unit used for the production control test shall be selected from the units passing the production inspection on the basis of one per every 25 units.

4. 6 Environmental tests. - Environmental tests shall consist of the following environmental tests of MIL-E-16400:

(a) Temperature.
(b) Humidity.
(c) Shock.

4. 6. 1 In addition to the tests specified in 4. 6, the energy dependency test (see 4. 8. 1) shall be performed.

4. 6. 2 Sampling plan. - The unit used for the environmental test shall be selected from the units passing the production inspection on the basis of one per every 50 units. The environmental tests may be performed concurrently with production control inspection on differenct samples.

4. 7 Disposition of shock tested equipment. - The shock tested sample shall be thoroughly reconditioned by the contractor before resubmitting it as production equipment.

4. 8 Test procedures.

4. 8. 1 Energy dependency. The equipment shall be tested to determine compliance with 3. 5. 1. 2, at 5 photon energies between 80 KEV and 1. 3 MEV. Cesium 137 and Cobalt 60 shall be two of the isotopes used to obtain these energies.

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4.8.2 Dose accuracy. - The equipment shall be tested to determine compliance with 3.5.1.3 by placing the equipment in a known field, using Cobalt 60 as the reference standard. The three points to be tested are 20 percent, 50 percent, and 80 percent of full scale.

4.8.2.1 Accuracy abbreviated. - The equipment shall be placed in a known field, using Cobalt 60 as the reference standard, until 40 roentgens are accumulated. The instruments shall read ± 20 percent of true dose.

4.8.3 High radiation rate independence. - At a dose rate of 10,000 r/hr., the recycling should not be less than 5 per second.

4.8.4 Leakage test. - The equipment shall be operated in background radiation for a period of 36 hours to determine compliance with 3.5.1.8.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Basic equipment or item. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. - Onboard repair parts shall be preserved, packaged, packed and marked as specified in MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation and packaging, packing and marking for shipment shall be as specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment which may be required when procurement is made (see 6.1).

5.2.1.1 Preservation, packaging, packing and marking. - The equipment shall be preserved, packaged, packed and marked in accordance with MIL-E 17555.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preservation, packaging, packing and marking requirements if other than those required by 5.1 (see 5.2).

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6.2 Preproduction. -

6.2.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 6665-N198SH)

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